

Marketing Performance Intelligence

Complete Power BI Project — Beginner to Portfolio Ready

Goal: Build a realistic multi-platform marketing analytics dashboard from raw data, clean it, model it, write DAX, design four dashboard pages, and turn the result into a portfolio project.

How to use this guide: Follow it in order. Do not skip ahead just to make the dashboard look pretty. The learning comes from understanding why every step exists.

Dataset: Marketing_Performance_Intelligence_Dataset.xlsx — includes FactMarketing, DimDate, DimPlatform, DimCampaign, DimContent, DimAudience, and MonthlyTargets.

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1. What We Are Building

Imagine you are a marketing analyst. A company is spending money on Instagram, YouTube, Facebook, LinkedIn and X. The manager does not want 30,000 rows of data. They want answers.

The main business question is:

“Given our marketing performance, where should we allocate our next ■10 lakh to maximize conversions and revenue?”

Your Power BI report will answer smaller questions that lead to that big answer:

- Which platform produces the most revenue?
- Which platform produces the best ROAS?
- Which campaigns waste money?
- Which content types create engagement?
- Which content types create conversions?
- Which audience segments are most valuable?
- Are we meeting monthly revenue and conversion targets?
- How has performance changed over time?

Final report structure

- Page 1 — Executive Overview
- Page 2 — Campaign Performance
- Page 3 — Content Intelligence
- Page 4 — Audience Analytics

Skills you will practice

- Power Query: importing, cleaning, data types, duplicates, missing values, transformations
- Data modeling: fact table, dimension tables, relationships, star schema
- DAX: measures, CALCULATE, DIVIDE, filter context, ranking and time intelligence
- Report design: slicers, KPI cards, charts, drill-through, tooltips and navigation
- Business analysis: turning numbers into recommendations

2. Understand the Dataset

Before opening Power BI, understand what each table means. Think of the model like a small company database.

Table	What it means	Important columns
FactMarketing	The main table containing marketing activity/performance	Date, IDs, Spend, Impressions, Reach, Engagements, Clicks, Leads, Conversions
DimDate	Calendar table used for dates and time analysis	Date, Year, Month, Quarter, Year_Month
DimPlatform	List of marketing platforms	Platform_ID, Platform
DimCampaign	List of campaigns	Campaign_ID, Campaign, Objective
DimContent	List of content formats	Content_ID, Content_Type
DimAudience	List of audience segments	Audience_ID, Age_Group, Gender, Audience_Segment
MonthlyTargets	Monthly business targets	Year_Month, Revenue_Target, Conversion_Target

The important idea: **FactMarketing** is where the numbers live. The Dim tables describe those numbers.

Expected relationships

```
DimDate[Date] 1 → * FactMarketing[Date] DimPlatform[Platform_ID] 1 → *
FactMarketing[Platform_ID] DimCampaign[Campaign_ID] 1 → * FactMarketing[Campaign_ID]
DimContent[Content_ID] 1 → * FactMarketing[Content_ID] DimAudience[Audience_ID] 1 → *
FactMarketing[Audience_ID]
```

The symbol **1 → *** means one row in the dimension can be connected to many rows in the fact table.

3. Phase 1 — Import the Data

You are currently at the Navigator window in Power BI. This is the exact screen you reached in our session.

Step-by-step

1. In the Navigator window, find the workbook called Marketing_Performance_Intelligence_Dataset.
2. Tick the checkbox beside DimAudience.
3. Tick DimCampaign.
4. Tick DimContent.
5. Tick DimDate.
6. Tick DimPlatform.
7. Tick FactMarketing.
8. Tick MonthlyTargets.
9. Do not press Load yet.
10. Press Transform Data at the bottom-right.

Why Transform Data instead of Load?

Load means “bring the data into the model now.” Transform Data means “open Power Query first so I can inspect and clean the data.” We deliberately want to clean first.

When Power Query opens, the left side should show seven queries. Click **FactMarketing**.

Checkpoint: You should see the FactMarketing table in the center and an Applied Steps panel on the right.

4. Phase 2 — Power Query Cleaning

This is where we make messy data trustworthy. The dataset intentionally contains a few problems so you can practice realistic cleaning.

4.1 Understand the preview

- The bottom may say “999+ rows”. That is only the preview; it does not mean the table has 999 rows.
- FactMarketing contains 30,030 rows when loaded from the generated file.
- There are 30 intentionally duplicated records.
- There are intentionally missing values in Reach, Clicks and Revenue.
- The Date values initially appear like numbers such as 45394 because Excel stores dates as serial numbers.

4.2 Remove duplicate records correctly

Important: Do not select only the Date column and remove duplicates. Many different marketing records can legitimately have the same date.

1. Click the small square in the top-left corner of the table, above the row numbers and to the left of Date. This selects the entire table.
2. Go to Home in the Power Query ribbon.
3. Choose Remove Rows.
4. Choose Remove Duplicates.
5. Look at Applied Steps on the right. You should see a new step called Removed Duplicates.
6. The fact table should now contain about 30,000 records.

4.3 Fix the Date column

Excel stores a date as a number of days from a starting date. That is why you may see 45394 instead of a normal date.

1. Click the Date column header.
2. Look at the small data-type icon beside the column name. If it says Whole Number or Any, that is not our final type.
3. Use Transform → Data Type → Date.
4. If Power Query asks for confirmation, choose Replace Current.

After this, the values should look like normal dates such as 2024-01-01 or 01/01/2024 depending on your regional settings.

4.4 Check data types

Column	Recommended type
Date	Date
Platform_ID / Campaign_ID / Content_ID / Audience_ID	Text
Spend / Revenue	Decimal Number

Impressions / Reach / Engagements / Clicks / Leads / Conversions	Whole Number
--	--------------

4.5 Missing values — do not blindly replace them

You will find missing values in Reach, Clicks and Revenue. A missing value is not automatically zero. Zero means the value is genuinely zero; blank means the value is unknown or unavailable.

For this learning project, keep the blanks in the fact table rather than replacing them with zero. Later, DAX measures such as DIVIDE will help us handle missing/zero denominators safely.

4.6 Clean the dimension tables

1. Click DimPlatform. Confirm Platform_ID is Text.
2. Click DimCampaign. Confirm Campaign_ID is Text.
3. Click DimContent. Confirm Content_ID is Text.
4. Click DimAudience. Confirm Audience_ID is Text.
5. Click DimDate. Confirm Date is Date and the calendar columns have sensible types.
6. Click MonthlyTargets. Confirm Year_Month is Text, Revenue_Target is Decimal Number and Conversion_Target is Whole Number.

4.7 Optional useful Power Query habit: rename steps

You can leave automatically generated step names, but learning to rename important steps makes complicated queries easier to understand. Right-click a step in Applied Steps → Rename.

5. Phase 3 — Build the Data Model

Click **Close & Apply** in Power Query. Power BI will load the cleaned data.

Go to Model view. You should see the tables as boxes.

Think of the model like this:

DimDate | | DimPlatform – FactMarketing – DimCampaign | | DimContent | | DimAudience

5.1 Create relationships

1. Drag DimDate[Date] onto FactMarketing[Date].
2. Drag DimPlatform[Platform_ID] onto FactMarketing[Platform_ID].
3. Drag DimCampaign[Campaign_ID] onto FactMarketing[Campaign_ID].
4. Drag DimContent[Content_ID] onto FactMarketing[Content_ID].
5. Drag DimAudience[Audience_ID] onto FactMarketing[Audience_ID].

For each relationship, you want:

- Cardinality: One to many (1:*).
- The dimension table is the “one” side.
- FactMarketing is the “many” side.
- Cross-filter direction: Single, from dimension to fact.

5.2 Why a star schema?

A star schema separates descriptive information from measurable events. It is easier to understand, usually performs better, and makes DAX behavior much easier to reason about.

Do not connect every table to every other table. Avoid unnecessary many-to-many relationships.

6. Phase 4 — Create and Mark a Proper Date Table

We already supplied DimDate, so you do not need to generate dates from scratch. However, you should learn why it exists.

A proper date table gives Power BI one continuous calendar that can support Year-to-Date, Previous Month, Year-over-Year and other time calculations.

6.1 Verify DimDate

1. Go to Data view and select DimDate.
2. Make sure Date contains every day from the beginning to the end of the dataset period.
3. Make sure Date is truly a Date data type.
4. Go to Table tools / Modeling tools and choose Mark as date table.
5. Select the Date column when Power BI asks which column identifies the date.

6.2 Sort Month names correctly

Alphabetical sorting would put April before August, which is wrong for a timeline.

1. Select DimDate[Month].
2. Choose Column tools → Sort by column.
3. Choose Month_Number.

For Year_Month, use the existing text column for display, but sort it using a proper chronological key if Power BI requires it. A numeric YearMonth key such as $\text{Year} * 100 + \text{Month_Number}$ is a common approach.

7. Phase 5 — Beginner DAX Measures

Now the fun part. DAX is the language Power BI uses for calculations. Start with simple measures and understand each one before moving to advanced formulas.

Create a dedicated table called **Measures** if you want a clean model. A simple way is Home → Enter Data → create a tiny table, then use it only as a home for measures.

Create these measures in the Measures table.

```
Total Spend = SUM(FactMarketing[Spend]) Total Revenue = SUM(FactMarketing[Revenue]) Total Impressions = SUM(FactMarketing[Impressions]) Total Reach = SUM(FactMarketing[Reach]) Total Engagements = SUM(FactMarketing[Engagements]) Total Clicks = SUM(FactMarketing[Clicks]) Total Leads = SUM(FactMarketing[Leads]) Total Conversions = SUM(FactMarketing[Conversions])
```

7.1 Why measures instead of calculated columns?

A measure calculates when the visual asks for it. If you put Platform on a chart, Total Revenue automatically recalculates for each platform. That is the power of filter context.

7.2 Ratio measures

```
Engagement Rate = DIVIDE([Total Engagements], [Total Reach]) CTR = DIVIDE([Total Clicks], [Total Impressions]) Lead Conversion Rate = DIVIDE([Total Leads], [Total Clicks]) Conversion Rate = DIVIDE([Total Conversions], [Total Clicks]) Cost per Lead = DIVIDE([Total Spend], [Total Leads]) Cost per Conversion = DIVIDE([Total Spend], [Total Conversions]) ROAS = DIVIDE([Total Revenue], [Total Spend])
```

Format Engagement Rate, CTR, Lead Conversion Rate and Conversion Rate as percentages. Format Spend, Revenue, CPL and Cost per Conversion as currency. Format ROAS as a decimal such as 3.42x.

8. Phase 6 — Intermediate & Advanced DAX

Once the basic measures work, build time intelligence and ranking. These are valuable interview skills.

8.1 Revenue growth

```
Revenue Previous Month = CALCULATE( [Total Revenue], DATEADD(DimDate[Date], -1, MONTH) )
Revenue MoM Growth % = DIVIDE( [Total Revenue] - [Revenue Previous Month], [Revenue Previous Month] )
Revenue Previous Year = CALCULATE( [Total Revenue], DATEADD(DimDate[Date], -1, YEAR) )
Revenue YoY Growth % = DIVIDE( [Total Revenue] - [Revenue Previous Year], [Revenue Previous Year] )
```

8.2 Running total

```
Running Revenue = CALCULATE( [Total Revenue], FILTER( ALLSELECTED(DimDate[Date]), DimDate[Date] <= MAX(DimDate[Date]) ) )
```

8.3 Ranking campaigns

```
Campaign Revenue Rank = RANKX( ALL(DimCampaign[Campaign]), [Total Revenue], , DESC, DENSE )
```

8.4 Revenue contribution

```
Revenue Contribution % = DIVIDE( [Total Revenue], CALCULATE( [Total Revenue], ALL(DimCampaign[Campaign]) ) )
```

8.5 Targets

Because MonthlyTargets is monthly rather than daily, you can either connect it through a Year_Month key or use a summarized target approach. Keep this separate from the main fact table so targets do not multiply when joined incorrectly.

9. Phase 7 — Executive Overview

This is the page a senior manager should understand in about 10 seconds.

9.1 Page layout

```
----- | Date Slicer | Platform Slicer |
Campaign Slicer | ----- | Spend | Revenue
| Conversions | ROAS | CTR | Conv Rate |
----- | Revenue & Spend Trend | Revenue by
Platform | | | | ----- | Campaign ROAS |
Spend vs Revenue | ----- | Top Campaigns /
Key Insight cards | -----
```

9.2 KPI cards

- Total Spend
- Total Revenue
- Total Conversions
- ROAS
- CTR
- Conversion Rate

9.3 Revenue and spend trend

1. Insert a Line and clustered column chart.
2. Put DimDate[Year_Month] on the X-axis.
3. Put Total Revenue and Total Spend into the values.
4. Sort Year_Month chronologically.
5. Use a slicer for Date so the user can change the reporting period.

9.4 Platform comparison

1. Insert a clustered bar chart.
2. Axis: DimPlatform[Platform].
3. Values: Total Revenue.
4. Add ROAS to the tooltip.
5. Sort descending by Revenue.

9.5 Spend vs Revenue scatter plot

Use Spend as X, Revenue as Y, Campaign as Details, and optionally Conversions as Size. This helps identify campaigns that spend a lot without producing proportional revenue.

9.6 Business insight box

Do not just display numbers. Add a text box titled **Management Takeaway**. Write one or two sentences after analyzing the visuals. Example structure: "Platform X generated the strongest ROAS, while Platform Y produced the largest reach. Reallocate incremental budget toward high-ROAS campaigns while monitoring scale."

10. Phase 8 — Campaign Performance

This page answers: **Which campaigns are working, and why?**

Recommended visuals

- Campaign leaderboard — Revenue
- Campaign leaderboard — ROAS
- Spend vs Revenue scatter
- Conversion funnel: Impressions → Clicks → Leads → Conversions
- Monthly campaign trend
- Campaign detail table

Campaign detail table columns

- Campaign
- Objective
- Spend
- Revenue
- ROAS
- Leads
- Conversions
- Cost per Conversion
- Conversion Rate
- Campaign Revenue Rank

Conditional formatting

Use conditional formatting to make strong and weak values obvious. For example, a data bar for Revenue and an icon or color scale for ROAS. Avoid turning the table into a rainbow; clarity is more important than decoration.

10.1 Decomposition Tree

1. Insert a Decomposition Tree.
2. Set Analyze to Total Revenue.
3. Add dimensions such as Platform, Campaign, Content Type and Audience Segment.
4. Click branches while exploring the report.
5. Use it to investigate why revenue is high or low.

The important learning: the Decomposition Tree lets the user interactively break a metric into dimensions instead of relying on a fixed chart.

11. Phase 9 — Content Intelligence

This page answers: **What type of content should our marketing team produce more of?**

Recommended analysis

- Content Type vs Engagement Rate
- Content Type vs CTR
- Content Type vs Conversion Rate
- Content Type vs Revenue
- Top content formats by platform
- Engagement by day of week
- Engagement by month

Important analytical lesson

Do not assume the content type with the highest engagement is the best content type overall. Engagement is only one stage of the funnel. A Reel might receive huge engagement while a Carousel might convert better.

A strong business conclusion might look like:

“Reels drive stronger engagement, but Carousels deliver better conversion efficiency. The recommended strategy is to use Reels for discovery and Carousels for conversion-focused campaigns.”

You must calculate the actual conclusion from the dashboard rather than copying this example.

12. Phase 10 — Audience Analytics

This page answers: **Who is valuable to the business?**

Recommended visuals

- Revenue by Age Group
- Conversions by Audience Segment
- Conversion Rate by Gender
- Spend vs Revenue by Audience Segment
- New/returning analysis if you extend the dataset later
- Audience performance matrix by Platform

Suggested slicers

- Platform
- Campaign
- Content Type
- Age Group
- Gender
- Date

Use drill-through

Create a Campaign Detail or Audience Detail drill-through page. The user can right-click a campaign in a table and jump to a page filtered to that campaign.

13. Drill-through, Tooltips, Bookmarks & Navigation

These features make the report feel like an application rather than a collection of charts.

13.1 Drill-through

1. Create a new page named Campaign Detail.
2. Drag DimCampaign[Campaign] into the Drill-through field well.
3. Add campaign-specific KPIs and charts.
4. Insert a Back button using Insert → Buttons → Back.
5. Go to a campaign chart on another page and right-click a campaign → Drill through → Campaign Detail.

13.2 Report page tooltip

1. Create a small tooltip page.
2. Set the page type to Tooltip.
3. Add a few useful metrics such as Revenue, Spend, ROAS and Conversions.
4. Turn Tooltip on for the relevant chart.
5. Hover over a campaign or platform to see the mini-report.

13.3 Page navigation

1. Insert buttons or a navigator.
2. Create navigation for Overview, Campaigns, Content and Audience.
3. Keep the navigation in the same position on every page.
4. Use consistent naming and spacing.

13.4 Bookmarks

Bookmarks can save a report state. A useful beginner example is a button that toggles between a Revenue view and a ROAS view, or shows/hides a filter panel.

14. Phase 11 — What-If Analysis & the ₹10 Lakh Decision

This is where the project becomes a business decision tool.

Create a What-If parameter for additional budget.

1. Go to Modeling → New parameter → Numeric range.
2. Name it Additional Budget.
3. Use a sensible range such as ₹0 to ₹1,000,000 with a step of ₹50,000.
4. Power BI will create a parameter and a measure.
5. Add the parameter as a slicer.

Then build a simple scenario measure.

$$\text{Scenario Revenue} = [\text{Total Revenue}] + ([\text{Additional Budget Value}] * [\text{Selected ROAS}])$$

The exact business formula can be improved later. The important lesson is that a parameter lets management change an assumption and see a different scenario.

A more mature recommendation

Instead of simply saying “put everything into the highest ROAS platform,” consider scale. A tiny campaign with fantastic ROAS may not be able to absorb ₹10 lakh. Your final recommendation should balance:

- ROAS
- Conversion volume
- Cost per conversion
- Historical consistency
- Campaign objective
- Platform scale
- Audience size

15. Phase 12 — Validation & Quality Checks

A professional dashboard is not finished when it looks good. It is finished when the numbers are trustworthy.

15.1 Basic checks

- Does Total Spend equal the cleaned fact-table total?
- Does Total Revenue equal the cleaned fact-table total, accounting for blanks?
- Are there duplicate dimension IDs?
- Does every FactMarketing foreign key have a matching dimension key?
- Does the Date relationship work?
- Do monthly charts sort correctly?
- Do slicers affect all expected visuals?
- Do percentages stay between sensible bounds?
- Does ROAS calculate correctly when Spend is zero or blank?

15.2 Relationship test

Put Platform in a table visual with Total Revenue. If each platform shows sensible values and the total equals the overall revenue, the platform relationship is probably working.

15.3 Filter-context test

Put Campaign in rows and Total Revenue in values. Add a Platform slicer. Select Instagram. The campaign revenues should change. If they do not, investigate the relationship.

16. Portfolio Documentation

When you finish, do not upload only a screenshot. Tell the story of the problem, your approach and the result.

Suggested README / case-study structure

- Business Problem
- Objective
- Dataset Description
- Data Cleaning
- Data Model
- DAX Measures
- Dashboard Pages
- Key Insights
- Recommendations
- Tools Used
- Future Improvements

Resume bullets

Use bullets like these, then replace placeholders with your actual findings:

- Built an interactive Power BI marketing intelligence dashboard analyzing 30K+ campaign records across five digital platforms.
- Designed a star-schema model with fact and dimension tables and created DAX measures for ROAS, CTR, conversion rate, CAC/CPC-style efficiency metrics, and time-based growth.
- Used Power Query to remove duplicates, standardize data types, validate missing values, and prepare analysis-ready datasets.
- Developed executive, campaign, content and audience views with drill-through, decomposition analysis, dynamic filtering and scenario planning.
- Translated dashboard findings into budget-allocation recommendations aimed at improving conversion efficiency and revenue.

LinkedIn project description

“Built a Marketing Performance Intelligence dashboard in Power BI to analyze spend, reach, engagement, leads, conversions and revenue across multiple digital platforms. The project covers Power Query data cleaning, star-schema modeling, DAX, time intelligence, campaign ranking, drill-through and what-if budget analysis.”

17. Interview Questions You Should Be Able to Answer

Practice explaining these in your own words. Do not memorize them like a script.

Why did you use a star schema?

Good answer: It separates descriptive dimensions from measurable fact data, keeps relationships simple and makes filtering and DAX easier to manage.

Why use measures instead of calculated columns for ROAS?

Good answer: ROAS needs to respond dynamically to the current filter context, so a measure is appropriate.

Why use DIVIDE instead of /?

Good answer: DIVIDE safely handles zero or blank denominators and lets you specify an alternate result.

What is filter context?

Good answer: It is the set of filters currently affecting a DAX calculation, coming from slicers, rows, columns, visual filters and relationships.

What does CALCULATE do?

Good answer: It evaluates an expression in a modified filter context.

What is the difference between a fact and dimension table?

Good answer: A fact table contains events/measures; dimensions describe those events and provide filtering/grouping attributes.

Why is the Date table important?

Good answer: Time-intelligence functions work reliably with a continuous, properly configured date table.

What is cardinality?

Good answer: It describes how many rows on one side relate to rows on the other side, such as one-to-many.

Why should dimensions filter the fact table?

Good answer: It keeps the model predictable and supports the standard star-schema pattern.

Why not replace all blanks with zero?

Good answer: Blank and zero have different meanings; replacing unknown values with zero can create false business conclusions.

How would you validate a dashboard?

Good answer: Reconcile totals to source data, test relationships, test filters, inspect edge cases and verify important calculations independently.

How would you decide where to spend another ■10 lakh?

Good answer: Compare ROAS, conversion efficiency, volume, consistency and scale, then test scenarios rather than relying on a single metric.

18. Final Checklist

Use this as your completion checklist.

- Imported all seven original tables
- Opened Power Query before loading
- Removed the 30 duplicate fact records
- Converted FactMarketing Date to Date type
- Checked every column's data type
- Reviewed missing values instead of blindly filling them
- Loaded the cleaned model
- Created one-to-many relationships
- Verified the Date relationship
- Marked DimDate as a date table
- Sorted Month by Month_Number
- Created basic DAX measures
- Created ratio measures using DIVIDE
- Created MoM and YoY growth measures
- Created campaign ranking
- Created revenue contribution
- Built Executive Overview
- Built Campaign Performance
- Built Content Intelligence
- Built Audience Analytics
- Added slicers
- Added drill-through
- Added tooltips
- Added navigation
- Added a What-If budget parameter
- Validated totals and filters
- Documented key business insights
- Prepared resume/portfolio description
- Practiced interview questions

Most important advice

Do not try to finish this in one sitting. Build one phase, understand it, test it, and then move to the next. If something breaks, that is not failure — debugging the model is part of learning Power BI.

When you get stuck, take a screenshot of the Power BI window and use this guide to identify the phase you are in. The next step should always be small and clear.